

UNITED STATES
PATENT AND TRADEMARK OFFICE



Real options in patenting: Role of secondary patent markets

Walter G. Park - American University

Gerard A. Torres - USPTO

Andrew A. Toole - USPTO

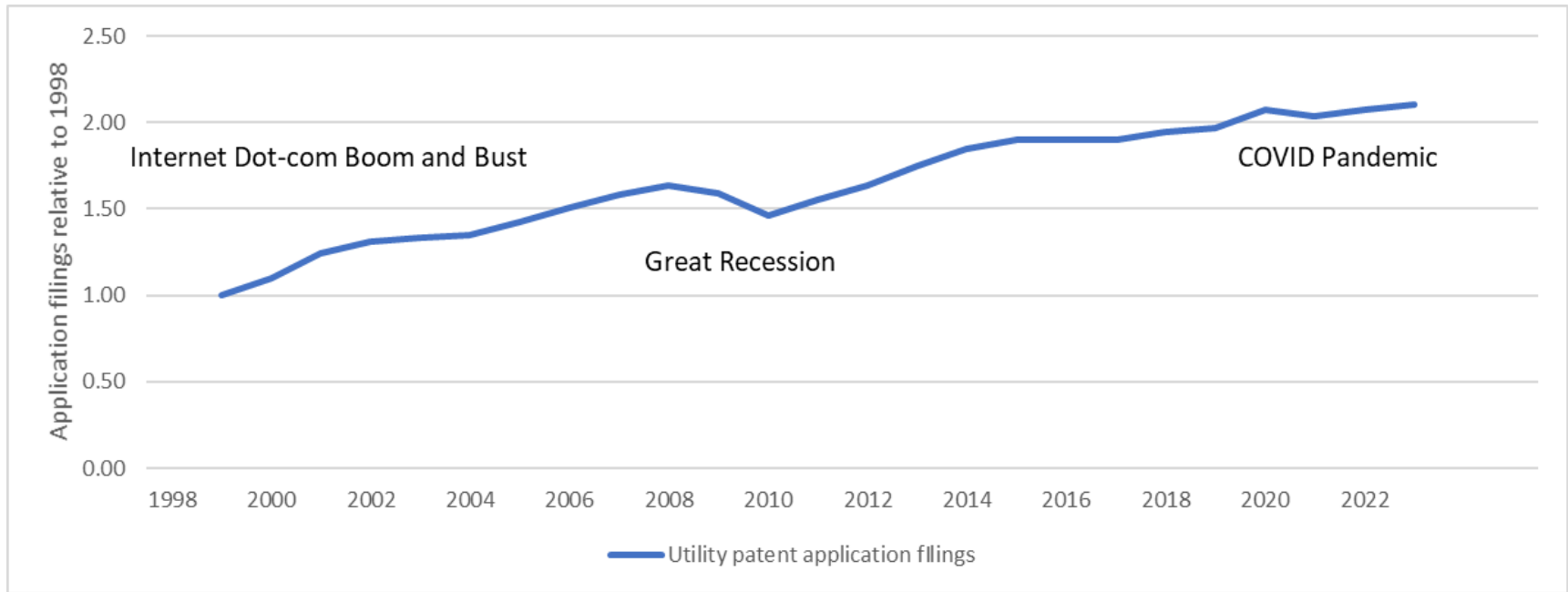
Ryan Hughes - USPTO

April 2024



UNITED STATES
PATENT AND TRADEMARK OFFICE ®

Economic upheaval is associated with strong swings in demand for patent protection



Notes: Utility patent applications filed by receipt date, date the application is submitted to USPTO.
Excludes requests for continued examination (RCE).

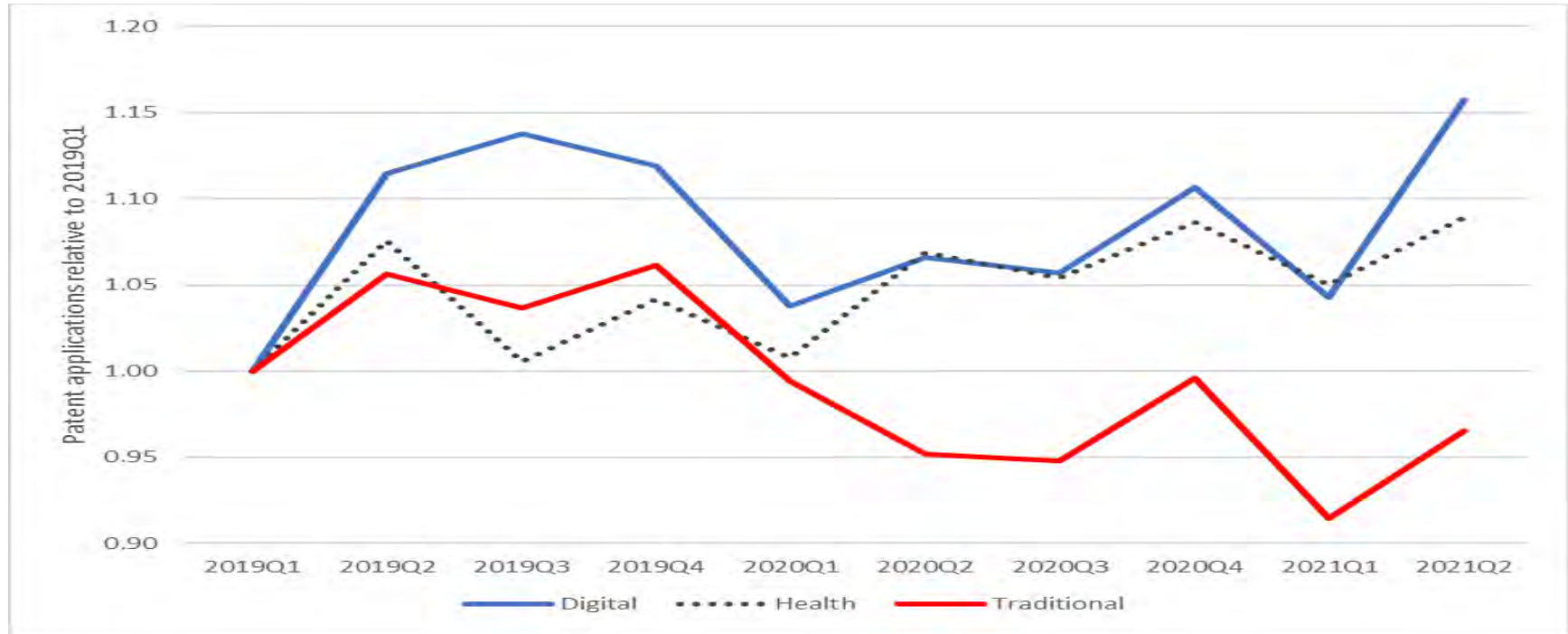
³ Source: USPTO Patent Application Locating and Monitoring (PALM) system.

Focusing-in on U.S. real gross output during Coronavirus pandemic



Notes: Seasonally adjusted gross real output by industry (chain-type price index).
Source: Authors' calculations based on U.S. Bureau of Economic Analysis.

U.S. patent filings by industry group during Coronavirus pandemic



Notes: Utility patent applications filed by receipt date, date the application is submitted to USPTO. The first-listed CPC symbol is used to classify each application. Origin counts are fractionalized based on each inventor's country code in their mailing address.

⁵ Source: Author's calculations based on data from USPTO's Patent Location and Monitoring system (PALM).



Research questions

Our purpose is to understand how economic shocks affect a firm's decision to seek patent protection.

We therefore ask the following questions:

- Do firms decrease their patenting activity amid economic shocks that create uncertainty of future financial returns?
- Do firms with sufficient access to secondary markets increase their patenting amid that same uncertainty?
- Can we gauge how much access to a secondary market is sufficient for a firm to switch from decreasing patenting activity to increasing it?

Real options framework

Developed for tangible investments ... but is applicable to intangible investments.

- Future patent applications can be expanded (i.e. filed sooner) or contracted (i.e. filed later).
- Future returns to patent applications may be uncertain.
 - When firms can resell patents at a later date, it possesses a *put option*.
 - When firms can acquire patents at a later date, it possesses a *call option*.
 - Higher values of the call option create incentives to postpone patenting, while higher values of the put option create incentives to patent currently.

How to assess firms' decisions?

Greater future uncertainty has an ambiguous effect on investments in patents, because it increases the value of both the *put* and *call* options.

Patent secondary markets allow innovation investments to be reversible. The more reversible, the greater the incentive to innovate and patent.

- For this study, reversibility is the option to sell or license patent rights to others.
- A firm can reverse, or recoup, their investment in patents.
- Patents are not a sunk cost and this affects a firm's decision to seek patent protection.

Models that measure uncertainty

1. Coefficient of variation (CV)
 - CV of Sales and Profits (Czarnitzki and Toole, 2011)
2. Auto-regressive (AR) model
 - Standard deviation of residuals from AR model on firm profit forecast (Ghosal and Loungani, 2000)
3. Stock Market volatility
 - Standard deviation of residuals of a firm's rate of return on stock price in relation to its industry and market index (Bloom and Van Reenan, 2000)
4. Cyclical-Trend filtering (Baxter-King Method)
 - Function of real sales per worker and cyclical component of real sales per worker (Hingley and Park, 2017)

Data sample

Unbalanced panel dataset (for exploration of data):

- Quarterly firm level data from 1990 – 2021.
- 1,800 companies (accounting for 40% of all U.S. grants).
- 15 countries represented, including U.S.
- 94 industries, North American Industry Classification System (NAICS) 4-digit level.

Balanced panel dataset (for final analysis):

- Quarterly firm level data from 1995-2018 to address censoring (ownership difficult to disambiguate until patent publication).
- 491 companies that operate in each year between 1995-2018.
- Required to address regression assumptions (see below).



Data sources

- *USPTO's PatentsView* – Granted patent applications
- *USPTO's PALM* – Abandoned patent applications (including unpublished)
- *Standard & Poor Capital IQ* – Firm financial data and name disambiguation
- *National Establishment Times-Series* – Firm name disambiguation
- *IAM market, Richardson-Oliver Insights (ROI)* – patent asset sales
- *USPTO's patent assignment records* – Patent licensing transactions
- *U.S. Bureau of Economic Analysis* – Business cycle data

Data preparation

- Firm name disambiguation
- Concordance: Match assignees/subsidiaries to parent companies
- Match patent application data to firm level data in Capital IQ
- Patent applications (both granted and abandoned) as of the date of filing.
- All relevant data converted to real 2012 dollars

Patent secondary markets

Using data on patent assets sold from ROI data and licensing transactions from USPTO's patent assignment data, we calculate a firm's secondary market exposure.

Assumptions:

- Some technology fields, as measured using the Common Patent Classification (CPC) system, are more involved than others in assets sales or licensing agreements.
- Firms apply for patents in a variety of technology fields.
- All of the CPC symbols on a patent application are necessary to analyze the full technological scope of the invention claimed (See: Section 905 of USPTO's Manual of Patent Examining Procedures).
- Patents are usually sold in packages, i.e. a group of patents are sold and purchased together.

Patent secondary markets (cont.)

The Index of Secondary Patent Market Sales Exposure (ISE) for firm i is:

$$ISE_i = \sum_{j=1}^N (Share_{ij} * Proportion_j)$$

Where:

- j indexes the technology field (CPC symbol), and N is the total number of technology fields the firm files in during a given sample period.
- $Share_{ij}$ is the share of the j^{th} technology field in the firm's overall patent applications.
- $Proportion_j$ is the share of the j^{th} technology field across all patent applications.
 - If $proportion_j$ is unweighted, we use raw counts: $\frac{CPC_i}{\sum_j CPC_j}$
 - If $proportion_j$ is weighted, we use the value of an imputed price (p) for field j : $\frac{p_j CPC_i}{\sum_j p_j CPC_j}$

Top CPC classes share of secondary market sales from ROI and licensing

Secondary Patent Market Sales

CPC Class Title	CPC Class	% share total
Electric Communication Technique	H04	43.29
Computing; Calculating; Counting	G06	27.20
Basic Electric Elements	H01	6.62
Measuring; Testing	G01	2.98
Basic Electronic Circuitry	H03	2.67
Information Storage	G11	2.07
Optics	G02	1.69
Signaling	G08	1.67
Medical or Veterinary Science	A61	1.52
Musical Instruments; Acoustics	G10	1.50
Electric Techniques not elsewhere classified	H05	1.01
Electric Power	H02	0.90

U.S. Licensing Transactions

CPC Class Title	CPC Class	% share total
Electric Communication Technique	H04	17.84
Medical or Veterinary Science	A61	14.41
Computing; Calculating; Counting	G06	13.27
Basic Electric Elements	H01	7.43
Engineering Elements	F16	4.34
Measuring; Testing	G01	3.71
Organic Chemistry	C07	3.33
Checking Devices	G07	2.54
Biochemistry	C12	1.82
Electric Power	H02	1.77
Land Vehicles other than Rails	B62	1.64
Vehicles in General	B60	1.57

Bolded are CPC subclasses that match in each market. Time period: 2009-2021 for secondary patent markets and 2000-2021 for licensing.



Estimating equation

Specification of model:

$$P_{it} = \exp(\beta_1 UNC_{it} + \beta_2(UNC_{it} * ISE_{it})) + X'_{it}\delta + \alpha_0 + \alpha_i + \alpha_j + \alpha_t)\varepsilon_{it}$$

- P_{it} – Number of applications filed at time t by firm i .
- UNC_{it} – The measure of uncertainty faced by firm i at time t .
- ISE_{it} – Index of exposure to secondary market/licensing market.
- X_{it} – Vector of time-varying firm controls: R&D, sales, cash flow, credit rating, number of employees, industrial concentration (HHI of sales share of industry).
- $\alpha_0, \alpha_i, \alpha_j$, and α_t are constant, firm, industry, and time fixed effects, respectively.
- ε_{it} is the stochastic error term.

Estimating equation (cont.)

Specification of model:

$$P_{it} = \exp(\beta_1 UNC_{it} + \beta_2(UNC_{it} X ISE_{it}) + X'_{it}\delta + \alpha_0 + \alpha_i + \alpha_j + \alpha_t)\varepsilon_{it}$$

Exponential model with Poisson estimator using pre-sample mean of P_{it} as firm fixed effect (Blundell et al. 1999; 2002).

- Addresses strict exogeneity.
- Newey-West standard errors robust to heteroskedasticity and serial correlation.

Hypothesis:

- $\beta_1 < 0$
- $\beta_2 > 0$

Results

- *A priori*, uncertainty has diverse potential effects, which can be associated with higher revenues or lower revenues (namely, the volatility).
- Our results consistently show $\beta_1 < 0$ and $\beta_2 > 0$.
 - Uncertainty has a negative association with patenting, but patenting increased for those firms with access to the secondary patent market.
 - Secondary market access varied by firm size and industrial sector.
- Investments in knowledge can be repurposed. They're not necessarily sunk costs.
- Patent assets are tradable (particularly in the digital technology fields).
- **Upshot:** study explains the resilience of some firms patenting under uncertainty.

Policy Implications

- An economic shock does not necessarily mean that there will be a decline in patent filings.
- The affect on patent filings from an economic shock is ambiguous, since firms may still continue to file patent applications if they are confident that there is a market for their innovations.
- While there is more knowledge and data on the market for patents than ever before, there is still much work to be done to better understand this market and how it impacts a firm's patent strategy.

Questions/Comments

Send questions or comments to:

Gerard Torres

Office of the Chief Economist

United States Patent and Trademark Office

gerard.torres@uspto.gov



